

Life on the Cusp: The Trident G Theory of General Intelligence

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November 28, 2025

Abstract

Trident G is a computational framework that reimagines both the free-energy principle and classic single-factor theories of g into a testable near-critical control model. Instead of understanding adaptive agents as pure free-energy minimisers that drive variational free energy F towards zero, Trident G treats the brain–body system as regulating a defended, strictly non-zero set point $F^\star$ within a viable operating band. Around this defended allostatic budget, a near-critical Ψ -band is defined over a cusp potential whose state axis is $x = F - F^\star$ and whose control axes are an engagement coordinate $\xi = G - F^\star$ (expected free energy) and an explore–exploit tilt δ . General intelligence is then reframed as a navigable $\mathcal{G}$ -volume: the portion of this Trident-shaped state–control space that a brain–body–niche ensemble can reliably reach, traverse, and re-enter without losing brain criticality benefits. I outline how this geometry can be instantiated by different physical carriers at micro (excitation–inhibition balance, acetylcholine, dopamine), meso (oscillations, long-range temporal correlations), and macro (salience networks, LC–NE, heart-rate variability) scales, and how niche coupling λ , meta-efficacy η , and coherence χ modulate Ψ -band occupancy. This framework gives us concrete, falsifiable predictions about Ψ -band time, joint EEG/HRV ' $F^\star$ corridors', salience–autonomic coupling, and ξ – δ -controlled reinforcement-learning agents. The aim is to inject some 'free energy' into the theoretical landscape on intelligent agents: to treat intelligence as the regulation of a viable near-critical band in a changing world, and to provide a generative scaffold for new experiments, analyses and AI architectures.

Keywords: *free-energy principle, near-criticality, general intelligence, allostasis, cusp catastrophe, cognitive resilience*

1. From Minimising Surprise to Regulating a Viable Band

The free-energy principle (FEP) casts the brain as a Helmholtz machine that minimises a variational free-energy functional F , which simultaneously upper-bounds sensory surprise and approximates the negative log-evidence of its generative model (Friston, 2010). In standard formulations this makes 'minimise F ' the default long-run objective: if the world is stable and the model class is fixed, reducing F is equivalent to becoming less surprised by data and more certain about latent causes. When conditions are slowly varying this is elegant. But when environments reconfigure on multiple timescales, pure minimisation results in fragile behaviour. Agents that insist on driving F towards zero are forced into chronically high-gain inference, incur high metabolic costs and struggle to forget obsolete representations (cf. Kanai et al., 2015). They become overfitted to past regularities and under-prepared for future regime shifts.

Trident G keeps this free-energy machinery but starts from a different control question. Rather than treating zero free energy as the goal, it treats the brain as maintaining a non-zero metabolic–accuracy trade-off at a defended set point $F^\star$. Here F is still variational free energy in the FEP sense – a composite of prediction error and model complexity – but, when coarse-grained over brain and body, it can also be read as a measure of allostatic load: the energetic and regulatory cost of keeping a particular generative model in play. $F^\star$ is a slowly tuned allostatic budget, greater than zero, that reflects both the organism's biological constraints and the typical demands of its niche.

On this view, the aim of control is not to eliminate surprise, but to keep deviations $F - F^\star$ within a viable operating band around this set point, so that the system can remain in a near-critical, flexible regime without sliding into chronic overload or metabolically expensive hyper-prediction. Formally, we write the instantaneous deviation as $x = F - F^\star$ and treat x as the state axis of the theory. Large positive x corresponds to sustained mismatch and stress; large negative x corresponds to over-regular, energetically costly hyper-prediction. Between these extremes lies a corridor of x -values in which the agent can think, learn, and act without damaging its long-term adaptability and performance. In the full computational

model (see OSF companion project for equations and code), this corridor is part of a Ψ -band of near-critical operation centred on $x \approx 0$ at a non-zero defended $F^\star$. The question for intelligence is then: how broad a range of this near-critical region can you stably occupy, and how reliably can you re-enter it after perturbations?

This reframing shifts what general intelligence is understood to be. Instead of a single scalar capacity, Trident G treats intelligence as the navigable volume $\mathcal{G}$ of a structured state–control space. $\mathcal{G}$ is the portion of a low-dimensional cusp landscape that an agent can stably visit and revisit while keeping x (the deviation of realised free energy/allostatic load F from the defended set point $F^\star$) within bounds. A large $\mathcal{G}$ means an agent can move between automated routines, flexible problem solving, and exploratory search while remaining adaptively regulated; a small $\mathcal{G}$ means a narrow repertoire confined to one or two safe wells.

In the rest of the paper, I make this concrete by introducing a simple cusp potential over x and two control dials: an engagement axis ξ (the expected deviation from the defended free-energy budget, i.e. $\xi = G - F^\star$) and an explore–exploit tilt δ that biases between alternative high-effort branches. In active inference, expected free energy G summarises the predicted mixture of epistemic and pragmatic value under a policy. In Trident G, $\xi = G - F^\star$ therefore acts as an engagement dial that tracks how far candidate policies are expected to push the agent away from its defended budget – from low-challenge, low-gain regimes to intense, high-variance regimes. This Trident-shaped geometry allows us to talk precisely about crystallised intelligence (Gc), fluid intelligence (Gf), and resilience (r) as different dimensions of control over the same landscape. Later sections formalise $\mathcal{G}$ in terms of control range, navigability, and niche coupling, with slow meta-variables such as meta-efficacy $\eta(t)$, coherence $\chi(t)$, and niche coupling $\lambda(t)$ tracking how coherently and how resiliently an agent’s generative model operates within its Ψ -band. For now, the key thesis is simple: intelligence is not bare surprise minimisation at a point, but viable-band regulation over a cusp defined by F , G , and a defended $F^\star$.

2. The Trident Cusp: Autopilot, Ψ -band, and Trident wells

To make this geometry precise, Trident G borrows a classic tool from catastrophe theory: the asymmetric cusp (Thom, 1975; Zeeman, 1976). The core proposal is that, at and above mesoscopic scales – from cortical hypercolumns to whole-brain control networks – cognition can be approximated by a potential V over a state axis x and two control parameters, engagement ξ and an explore–exploit tilt δ . Up to a linear rescaling of x and ξ , and with slow meta-states $(\eta, \chi, \lambda, F\star)$ held fixed, we can work with the normalised form:

$$V(x; \xi, \delta) = x^4 - \xi x^2 - \delta x$$

Here $x = F - F\star$ is the instantaneous deviation of realised free energy from the defended budget, $\xi = G - F\star$ is the corresponding deviation in expected free energy G under the current generative model (subjectively, engagement or expected challenge), and δ is an effective bias towards convergent ($\delta > 0$) versus divergent ($\delta < 0$) inference. In the full computational model, this time-varying tilt $\delta(t)$ is a fast, state-dependent tilt driven by valence-weighted prediction error and modulated by coherence $\chi(t)$ and niche coupling $\lambda(t)$ (formal equations and simulations in the OSF project). For the purposes of this opinion piece we treat δ as a control dial that selects between explore- and exploit-dominated branches of high-effort processing, and absorb the influence of η, χ and λ into the effective shape of the cusp.

For fixed (ξ, δ) , gradient descent on V pulls the system towards one or more stable minima in x . As ξ and δ vary, the number and depth of these minima change in a characteristic way. Figure 1 summarises the resulting Trident geometry. Panel A shows the equilibrium surface in (ξ, δ, x) for the normalised potential, with darker points marking a Ψ -band collar around the cusp where barriers are shallow and switching between attractors is easy. Panel B shows the corresponding potential surface $V(x; \xi, 0)$ at zero tilt, illustrating how a single Autopilot basin for $\xi < 0$ flattens into a near-critical ridge as ξ approaches 0 and then splits into two outer 'bowls' for $\xi > 0$. Panel C presents one-dimensional slices $V(x; \xi, 0)$ for representative values of ξ , making visible the transition from a single low- $|x|$ basin (Autopilot /

Gc), through a shallow, stretched Ψ corridor, to two high- $|x|$ wells (Gf explore and exploit branches).

Panel D projects the equilibrium states into the (ξ, x) plane at $\delta = 0$, producing the Trident shape: a central Autopilot shaft near $x \approx 0$, capped by a Ψ -band collar, and two symmetric Gf 'prongs' at higher $|x|$.

[Insert Figure 1 about here.]

This geometry gives three qualitatively distinct computational regimes. For sufficiently negative engagement ($\xi < 0$), the landscape has a single deep well centred near $x \approx 0$. This is the crystallised-intelligence regime: well-learned habits and skills operate within a stable attractor state. Barrier heights are high, so the system resists leaving this basin. Behaviourally this resembles habitual control in cortico-striatal loops, where policies are low-cost, reliable and resistant to modification (Keramati et al., 2011; Yin & Knowlton, 2006). Consistent with this, Default Mode Network (DMN) activity and connectivity increase during the automated application of learned rules, and default-mode–medial-temporal interactions predict faster, memory-based 'autopilot' responses under stable task contexts (Vatansever et al., 2017), suggesting that a DMN-centred associative workspace supports these low-engagement Gc regimes.

As engagement rises towards intermediate values ($\xi \approx 0$, small $|\delta|$), the landscape flattens into a shallow bifurcation with a central ridge. This is the Ψ -band. Because a Griffiths-phase-like heterogeneity produces a collar of near-critical dynamics around the nominal cusp, the upper part of the Autopilot shaft and the upper flanks of the Trident wells already contain patches with Ψ -like dynamics that Gf -related operations can recruit when needed (Moretti & Muñoz, 2013). Barriers between sub-basins are low, drift dominates, and the system can metastably explore the rim of the cusp. This is a g -relevant regime: multiple partial options can be maintained simultaneously, combined and pruned without the cost of fully committing. At the network level, externally oriented fluid reasoning in this region is implemented mainly by the Fronto-Parietal Control Network B (FPCN-B) in concert with the Dorsal Attention Network (DAN), which rapidly reconfigures task rules and shifts attention between shallow sub-basins, while

deeper relational map operations and long-horizon strategy formation are carried by hippocampus–orbitofrontal maps coupled to the default-mode network and the more internally oriented FPCN-A (Dixon et al., 2018). In Trident G terms, FPCN-B provides the capacity 'cursor' that moves along a given cusp landscape, whereas FPCN-A slowly remaps what ξ , δ and the wells mean in a domain by reshaping the underlying successor-style maps and schemas.

For higher engagement (ξ beyond a small positive threshold ε), the landscape becomes a pronounced W-shaped double well. The tilt parameter δ now matters. Positive δ deepens one well, favouring convergent reasoning: analytic, goal-directed exploitation of a chosen model. Negative δ deepens the other, favouring divergent reasoning: exploratory, generative search through alternatives. In this regime the system is fully engaged and large-scale reconfiguration is possible, but also risky if barrier heights become too high or if the system is repeatedly forced into one branch without opportunities to return via the Ψ -band collar.

The cusp rim near $\xi \approx \varepsilon$ marks a commitment threshold: as engagement drifts across this boundary, the system leaves the shallow Ψ corridor and falls more deeply into one of the two basins, favouring either exploitation or exploration. The barrier between these basins acts as a state-dependent switching cost: it is low near the Ψ -band, allowing rapid shuttling between options, but becomes higher when training, stress, or chronic niche capture deepen a particular basin, producing hysteresis in sequential choice. This offers a simple geometric way to think about path dependence, habit formation, and getting stuck in a particular cognitive mode, and provides a natural target for interventions that deliberately lower barriers near the Ψ rim.

In what follows, I use 'g-loop' as shorthand for a single episode-long trajectory of this control system through the Trident cusp that is Ψ -dominated: an Autopilot-collar $\rightarrow \Psi \rightarrow$ Trident-collar $\rightarrow Gc$ pass, in whole or in part. The g-loop labels such as Autopilot, Boredom Reset, Problem Engaged, Diverge, Converge, Solution, and Consolidate – should not be read as discrete boxes on a diagram but as

regimes along continuous trajectories in the (x, ζ, δ) space. A typical successful episode might trace a full loop (Gc shaft $\rightarrow \Psi$ (divergent / explore) $\rightarrow \Psi$ (convergent / exploit) $\rightarrow Gc$), but many are truncated (a brief Ψ excursion then back to Autopilot) or spend time shuttling between exploit and explore collars before finally consolidating. Episodes that are captured deep in the Autopilot shaft or in a single Trident well correspond to broken or dysregulated loops rather than healthy g-loops.

At what physical scale does this potential live? A single perfectly homogeneous microcircuit can show a local excitation–inhibition-tuned bifurcation, but it is too limited to support the kind of stretched criticality that intelligence requires. Trident G therefore treats a hypercolumn or small cortical patch ($\approx 10^4$ – 10^5 neurons) as the minimal 'cusp unit'. Within such a patch, heterogeneity in local excitation–inhibition ratios, dendritic trees, and interneuron types produces a spread of effective ζ and gains; shared neuromodulatory input supplies the common δ tilt. A Griffiths phase of many such units – each with slightly different local parameters – naturally yields a Ψ -like region in which a sizeable fraction of units are near-critical at any time, generating power-law avalanche statistics with continuously varying exponents (Moretti & Muñoz, 2013). The cusp potential above should therefore be read as the mesoscopic envelope of this heterogeneous 'stack'.

In this picture the engagement and tilt 'dials' (ζ, δ) are not freely set by a hidden homunculus but by a salience–allostasis controller anchored in the Saliency Network (SN) and its tight coupling to autonomic and endocrine systems. The anterior insula and dorsal anterior cingulate, which form the cortical core of the SN, integrate exteroceptive surprise with interoceptive error and body-budget status, and project densely to brainstem and hypothalamic nuclei that regulate sympathetic tone, vagal output, and hypothalamic–pituitary–adrenal (HPA)-axis activity (Critchley & Harrison, 2013; McEwen & Gianaros, 2010; Seeley et al., 2007). On the present account this salience–allostasis loop reads the composite deviation $x = F - F^\star$, generates fast arousal and 'need-to-act' signals, and adjusts ζ via locus-coeruleus–noradrenaline gain and δ via dopaminergic bias, while slower statistics such as χ , η , and λ reshape its gains and effective barrier heights over longer timescales (Menon, 2011). When trajectories

are repeatedly driven far outside the Ψ -band – either into chronically low-engagement Autopilot or into highly stressed Trident wells – the system loses near-critical dynamics, Ψ -occupancy drops, η decays, and regulation fails. Agency in Trident G is therefore an emergent property of this nested control architecture and its ability to keep g-loops Ψ -dominated across diverse tasks, not a separate inner agent adjusting parameters.

Within this envelope, general intelligence is the viable parameter space $\mathcal{G}$: the range of (x, ξ, δ) values for which an agent can sustain Ψ -dominated g-loops without losing viability. Later sections unpack how $\mathcal{G}$ factors into control range, navigability, and niche coupling, and how classic constructs such as Gf and resilience r correspond to orthogonal slices through this Trident geometry. For present purposes, the key point is that a simple cusp over a defended set point $F\star$ yields Autopilot, flow, exploitation, exploration, their hysteretic transitions, and their breakdown under loss of Ψ -occupancy in a single picture.

3. Micro-Scale Evidence

Having introduced the Trident cusp as an abstract control geometry, we can now ask whether its variables (x, ξ, δ) have plausible carriers at the level of synapses and cortical columns.

The g-Loop at the Cortical Hypercolumn Level

At the hypercolumn / minicolumn ($\approx 10^4$ – 10^5 neurons) scale, Trident G treats $x = F - F\star$ as a local deviation signal that tracks how 'energised' a microcircuit is relative to its defended activity budget: for example, differences between superficial L2/3 pyramidal burst rates and deeper feedback rates, or more generally any operational proxy for excitation–inhibition balance and recurrent drive. The engagement coordinate $\xi = G - F\star$ is read as a local gain or precision signal carried by basal forebrain acetylcholine (ACh) and allied neuromodulators that tune the expected impact of inputs and recurrent predictions. The explore–exploit tilt $\delta(t)$ is implemented, at least in part, by opponent D1/D2 signalling in

cortico-striatal microcircuits, which biases updates towards 'go / exploit' versus 'no-go / explore' policies on that patch.

On this view, the cusp manifests at the micro-scale as a neuromodulator-gated pitchfork bifurcation with a dopaminergic tilt. Step-like increases in ACh are hypothesised to push the local excitation–inhibition ratio through a critical region, flattening a single firing-rate mode and, beyond a threshold, splitting activity into two quasi-stable high- and low-activity branches separated by an unstable regime – the local signature of ξ crossing zero into the Ψ -band (cf. critical-like phenomena in cortical cultures; Haldeman & Beggs, 2005). Dopamine is hypothesised to provide the tilt: bursts that transiently favour D1-mediated 'go' synapses correspond to $\delta(t) > 0$ (convergent, exploit-biased trajectories), whereas D2-dominated states correspond to $\delta(t) < 0$ (more divergent, explore-biased trajectories), in line with standard models of cortico-striatal learning (Cools, 2019; Yin & Knowlton, 2006).

In the full computational model, these neuromodulator dynamics enter not as additional axes but as fast, state-dependent influences on $\xi(t)$ and $\delta(t)$, with their slower statistics feeding into the meta-states η , χ , and λ . For present purposes it is enough to note that ACh and dopamine are plausible micro-carriers of the Trident cusp: they tune local x and ξ , and provide a time-varying $\delta(t)$ tilt. The OSF companion project spells out the mapping from neuromodulator concentrations to cusp parameters in more detail and lists concrete *in vitro* predictions, such as the emergence of bimodal firing-rate distributions as ACh is swept across a critical band.

4. Meso/Macro Human Data

The same control roles also appear at larger scales, where oscillatory rhythms, large-scale brain networks, and autonomic signals provide natural carriers for deviation (x), engagement (ξ), and tilt ($\delta(t)$).

The g-Loop in ECoG, EEG/MEG, and fMRI

Table 1

Candidate meso- and macro-scale proxies for the Trident G variables. These are not unique identifiers, but concrete ways to approximate how far the system is from its defended near-critical corridor, how strongly it is engaged with a task, and whether it is momentarily tilted toward exploitation or exploration.

Scale	Candidate carrier for $x = F - F^\star$	Candidate carrier for $\zeta = G - F^\star$	Candidate carrier for $\delta(t)$	Example empirical signature
Meso (cortical patch; ECoG/EEG/MEG)	Combination of local β -band (15–30 Hz) power and alpha-band long-range temporal correlations (α -LRTC exponent) as proxies for local E:I balance and distance from criticality	Task-locked changes in alpha phase coherence between fronto-parietal and sensory regions (e.g., pulvinar/precuneus gating)	Aperiodic $1/f$ γ -slope and related spectral tilt as proxies for the momentary explore–exploit bias $\delta(t)$	Prediction: when alpha coherence drops below a threshold, β -power distributions become bimodal (Autopilot vs Trident states), and $1/f$ γ -slope biases predict exploit vs explore choices on the next trial
Macro (whole-brain networks + physiology)	Joint position of global BOLD signal variability σ_c and the short-term fractal exponent α_1 of heart-rate variability as a brain–body 'distance from $F^\star$ '	Baseline pupil diameter and task-locked changes in network integration/segregation as markers of expected challenge and arousal	PET-based D1/D2 receptor occupancy ratio and behavioural indices of exploration vs exploitation as slow and fast correlates of $\delta(t)$	Intermediate σ_c and α_1 values associated with flexible switching between networks and subjective 'flow'; pharmacological or stress manipulations shift these distributions

At the mesoscopic level, invasive ECoG and scalp EEG/MEG studies show that alpha-band activity exhibits long-range temporal correlations (α -LRTCs) that are strongest in an intermediate regime and become either weakened or pathologically rigid when systems move away from critical-like dynamics (Hardstone et al., 2012; Linkenkaer-Hansen et al., 2001). In the Trident G framing, these α -LRTCs provide a mesoscopic marker of staying near the Ψ -band: exponents near 0.5 indicate subcritical,

noise-like activity, exponents approaching 1 indicate over-stable, supercritical structure, and an intermediate band corresponds to the defended near-critical corridor.

Within this corridor, Trident G predicts that task engagement will modulate alpha coherence and beta power in a way that reflects the cusp geometry. When alpha coherence between fronto-parietal and sensory regions drops below a critical threshold, local circuits are hypothesised to leave the Autopilot shaft and enter the Trident regime, so that β -band power distributions become bimodal, reflecting switches between low- and high-engagement attractor states. The aperiodic $1/f$ γ -slope of the local power spectrum is a natural candidate proxy for the momentary tilt $\delta(t)$: a positive γ -slope would bias towards convergent, exploit-like choices, whereas a negative γ -slope would bias towards divergent, explore-like choices (Gao et al., 2017; Lendner et al., 2020; Shine, 2019). These patterns are treated as testable predictions rather than established facts, and the OSF companion project spells out specific ECoG/MEG analyses that could probe them.

At the systems level, this cortical cusp dynamics appears to be coupled to a salience–allostasis controller in which the insula–anterior cingulate salience network drives locus coeruleus–noradrenaline arousal and autonomic / endocrine responses. Baseline pupil diameter and tonic locus coeruleus firing track shifts between exploitation and exploration in decision tasks, consistent with adaptive gain accounts of LC–NE function, and the same salience hub projects to hypothalamus and brainstem nuclei that implement body-budget adjustments (Aston-Jones & Cohen, 2005; Breton-Provencher & Sur, 2019; Critchley & Harrison, 2013; Jepma & Nieuwenhuis, 2011; Menon, 2011; Seeley et al., 2007). In Trident G terms, this loop tracks $x = F - F^\star$ at the brain–body level and adjusts ξ and $\delta(t)$ accordingly.

Whole-brain fMRI suggests a similar geometry at the macroscale. During resting state, the standard deviation of the global signal, σ_c , often varies in ways that track how flexibly networks reconfigure (Deco & Kringelbach, 2016). On the present account, intermediate σ_c values correspond to Ψ -band operation, whereas very low or very high σ_c reflect subcritical and supercritical regimes.

Dopaminergic manipulations are expected to shift the distribution of ζ and the effective tilt $\delta(t)$, with D1/D2 occupancy ratios providing slow, trait-like constraints and trial-by-trial prediction-error signals providing fast, state-dependent tilts.

Operationally, $F^\star$ can be approximated as the regime where cortical long-range temporal correlations and cardiac fractal dynamics both occupy an intermediate, near-critical range (Linkenkaer-Hansen et al., 2001; Peng et al., 1995). In practice, this can be estimated using detrended fluctuation analysis on band-limited EEG or MEG power to obtain a cortical LRTC exponent, together with the short-term fractal exponent α_1 of heart-rate variability (Goldberger et al., 2002; Hardstone et al., 2012; Iyengar et al., 1996). The defended set-point $F^\star$ would then correspond to the band of task and arousal states in which these exponents place an individual 'in the zone': behavioural performance is high, subjective effort is sustainable, and allostatic load remains moderate (Palva et al., 2013; Thayer et al., 2012).

On the Trident G view, vertical deviations $x = F - F^\star$ index the degree of displacement from this joint brain–body near-critical corridor, and resilience reflects how reliably and how quickly trajectories return, as signalled by salience-network–autonomic coupling (Critchley & Harrison, 2013; McEwen & Gianaros, 2010). This makes the mind–body near-critical corridor a concrete target for neurofeedback and training in later applications, for example by using real-time EEG/MEG and HRV feedback to help agents inhabit Ψ -dominated g-loops more often (Ros et al., 2014).

5. Implications: Niches, Pathology and AI Agents

Trident G makes intelligence explicitly relational by insisting that agents are evaluated not only by how they move on the cusp, but by how their actions couple back into their environmental or cognitive niche (Hohol, 2021; Pinker, 2010). The computational model summarises this with a niche-coupling meta-state $\lambda(t)$. When λ is in a healthy, moderate range, the environment provides structured but non-adversarial feedback: prediction errors are informative, policies can change the niche in graded ways,

and the defended Ψ -band and its associated $F\star$ corridor are preserved. In this regime, ξ remains a meaningful engagement dial rather than being driven purely by noise or by externally imposed constraints.

By contrast, captured or impoverished niches either anti-correlate with the agent's intentions or over-predict its state. Rigid hierarchies, filter-bubble media, or highly constrained work environments can all drive λ toward zero by cutting off controllable feedback and collapsing the range of actions that matter (Shew & Plenz, 2013; Srinivasa et al., 2015). In the Trident G geometry this flattens the cusp and narrows $\mathcal{G}$: either the system retreats into a single Autopilot basin, repeating a few safe routines, or it is pushed into chronically stressed Trident wells without a reliable route back through the Ψ -band. Enriched animal environments, where λ is higher and surprise remains structured but not overwhelming, appear to keep neural avalanches near Griffiths-like exponents and wide dynamic range, consistent with sustained near-critical operation (Haldeman & Beggs, 2005; Moretti & Muñoz, 2013). In human organisations, loosely coupled, semi-autonomous teams resemble small $\lambda > 0$ near-critical 'cells' that can explore and stabilise new patterns, whereas large, tightly controlled organisations behave more like captured wells.

The same geometry offers a compact way to sketch common psychiatric pictures as distortions of ξ , $\delta(t)$, η , and λ , rather than as entirely separate mechanisms. In attention-deficit/hyperactivity disorder (ADHD), salience and arousal systems appear to switch too quickly and inconsistently, with aberrant coupling between salience and control networks and unstable engagement with tasks (Castellanos & Proal, 2012; Menon, 2011). On a Trident G view, ξ schedules are too volatile and η decays quickly: the system is frequently displaced out of the Ψ -band before task-relevant structure can stabilise, producing distractibility and impulsive switching between shallow attractors.

In obsessive-compulsive disorder (OCD), cortico-striatal loops show over-stable habit circuits and a bias toward over-checking and over-control (Cools, 2019; Gillan et al., 2016). Geometrically, this looks less like a global $\delta > 0$ trait and more like certain loops having deep, narrow Autopilot wells with

high barriers: once ξ is captured in these convergent routines, it struggles to return to the Ψ ridge or to recruit alternative Trident branches. In major depression, chronic stress and allostatic load are associated with reduced variability in large-scale networks, blunted reward responsivity, and reduced heart-rate-variability indices of flexible regulation (Goldstein & McEwen, 2002; McEwen, 1998; Thayer & Lane, 2000). Here ξ often hovers near zero, η is low, and λ is effectively weakened: the geometry shrinks toward a single shallow basin, so the system can neither recruit deep Trident modes nor reliably re-enter a healthy Ψ -band.

Conversely, manic or addictive states can be read as supercritical failures: the system becomes confined to a deeply engaged Trident basin with extreme $|\delta(t)|$ and elevated ξ , repeatedly exploiting or exploring along a narrow behavioural groove. Short-term η may be high within that groove – actions feel effective – but Ψ -occupancy collapses and recovery back to the defended $F\star$ corridor is impaired. Taken together, these sketches illustrate both subcritical (collapsed, low-variability) and supercritical (over-engaged, high-variability) distortions of the same cusp geometry, and suggest that restoring healthy Ψ -band occupancy and λ may be a common therapeutic target.

The Trident G control framework also suggests a practical modification of reinforcement-learning and active-inference agents. Standard agents are typically optimised to minimise expected free energy or maximise cumulative reward, and often converge on narrow, brittle policies that forget earlier tasks when objectives change (Henderson et al., 2018; Khetarpal et al., 2022; Kirkpatrick et al., 2017). From a systems perspective, a relentless drive to minimise loss encourages over-commitment: policies catastrophically forget when task contingencies shift, modes collapse into a small set of hyper-optimised behaviours, reward hacking appears with no graceful degradation route, and compute is wasted on maintaining high precision everywhere rather than defending a limited 'metabolic' budget.

A Trident G-inspired controller instead treats ξ and $\delta(t)$ as first-class variables and makes flexibility a primary control objective rather than a post hoc regularisation term. Plasticity is suppressed in

deep Autopilot or highly engaged Trident basins and concentrated on the Ψ -band where multiple options remain viable and barrier heights are low. In reinforcement-learning terms, this means maintaining a predictive 'successor' map of future state occupancies, rather than only caching a single scalar value function tied to a particular reward schedule. This is analogous to successor-representation and options frameworks, which explicitly separate transition structure from value and reuse predictive maps across tasks (Barreto et al., 2017; Gershman, 2018; Momennejad et al., 2017; Stachenfeld et al., 2017). Trident G adds to these frameworks an explicit near-critical Ψ -band and an engage/tilt controller that decides when to recruit extra control, similar in spirit to expected-value-of-control accounts of anterior cingulate function (Shenhav et al., 2013, 2017).

In non-stationary benchmarks, such agents are predicted to learn faster and show less catastrophic interference than reward-only baselines, especially when tasks share latent structure. Within the full computational model, this corresponds to explicitly tracking Ψ -occupancy, η , and λ , and treating the navigable $\mathcal{G}$ -volume – not just expected return – as an optimisation target. For interactive systems such as dialogue models, the same principle would amount to defending a Ψ -band in which the agent can pivot between script-following (convergent) and creative recovery (divergent) as conversational entropy and uncertainty move ξ and $\delta(t)$, rather than single-mindedly driving prediction error to zero. The OSF companion project sketches concrete RL architectures that instantiate this geometry and suggests benchmark tasks where they can be tested against standard baselines.

6. Falsifiable Predictions

Trident G is intended as a framework for generating new, testable hypotheses through novel conceptual synthesis. The cusp geometry, the Ψ -band, and the defended $F\star$ corridor are only useful if they pick out measurable regularities that go beyond existing 'capacity-only' accounts of intelligence and resilience. Here I sketch a non-exhaustive set of predictions that follow from the current model. The OSF

companion project provides more detailed task designs, simulation code, and suggested analysis pipelines for each class of test.

Ψ -Band Occupancy and Performance

If the near-critical Ψ -band is where flexible fluid reasoning and healthy regulation are concentrated, then time spent in this band should matter over and above raw capacity. Trident G predicts that individual differences in Ψ -occupancy – indexed by joint signatures such as beta-power bimodality, intermediate $1/f$ slopes, and Griffiths-like avalanche exponents – will correlate with both fluid intelligence and behavioural / physiological resilience beyond variance explained by traditional capacity measures (e.g., working-memory span, processing speed) (Beggs & Plenz, 2003; Cocchi et al., 2017; Hardstone et al., 2012; Moretti & Muñoz, 2013). Agents or individuals with larger estimated $\mathcal{G}$ -volume should show (a) broader ranges of near-critical exponents at rest and (b) shorter re-entry times into the Ψ -band after perturbation (for example, following acute stress or cognitive load).

Mind–Body $F\star$ and Resilience

On the Trident G account, each individual has a reproducible $F\star$ corridor – the individualised slice of the Ψ -band defined as a joint optimal range for cortical long-range temporal correlations and HRV α_1 fractal exponents – in which performance is high and allostatic load is moderate.. Deviations $x = F - F\star$ away from this corridor should predict both subjective effort and reduced accuracy, and resilience should be reflected in how reliably and how quickly LRTC and HRV indices return to this band after stress or fatigue. More resilient individuals, with higher effective r , are predicted to show (a) a wider safe band in this joint EEG / HRV space and (b) more frequent spontaneous returns to it across days (Goldberger et al., 2002; Iyengar et al., 1996; Palva et al., 2013; Peng et al., 1995; Thayer et al., 2012). In longitudinal data, the width and occupancy of this $F\star$ corridor should prospectively predict vulnerability to burnout and mood disorders.

Salience–Allostasis Control and ξ – η Dynamics

If the salience network and its projections to LC–NE and autonomic / endocrine targets implement the ξ / η controller, then individual differences in salience-network integrity and SN–ANS

coupling should predict the dynamic range of ζ and the long-run meta-efficacy η (Aston-Jones & Cohen, 2005; Breton-Provencher & Sur, 2019; Critchley & Harrison, 2013; Menon, 2011; Seeley et al., 2007; Thayer et al., 2012). Specifically, stronger insula–ACC connectivity and more flexible LC–NE phasic / tonic patterns should be associated with (a) more precise placement of the system in Autopilot, Ψ , or ‘trident’ zones on demand and (b) faster, more reliable re-entry into the Ψ -band after large $|x|$ excursions. Conversely, weakened or rigid SN–ANS coupling should predict shallower ζ trajectories, lower η , and greater time spent in subcritical (collapsed) or supercritical (over-engaged) regimes.

RL / AI Agents with ζ – δ Control

Reinforcement-learning or active-inference agents that explicitly track ζ and $\delta(t)$, concentrate plasticity on the Ψ -band, and quench it in deep Autopilot or Trident wells should display a different dynamical and behavioural profile from otherwise similar agents that only minimise free energy or maximise reward (Barreto et al., 2017; Gershman, 2018; Momennejad et al., 2017; Ramstead et al., 2023; Srinivasa et al., 2015; Stachenfeld et al., 2017). Dynamically, such ζ – δ -controlled agents are predicted to show Griffiths-like resting activity (power-law avalanche statistics with exponents in the range reported for near-critical systems) rather than quickly collapsing into a single attractor. Functionally, they should be more robust to non-stationarity and adversarial or shifted inputs, retaining earlier tasks better when goals are swapped and showing smoother degradation rather than catastrophic interference. This would support the theory’s claim that intelligence is best understood as regulation of a navigable band in state–control space, rather than as pure loss minimisation. The OSF companion project outlines benchmark suites and ablation tests to evaluate these predictions.

7. Conclusion: g as Navigable $\mathcal{G}$ -Volume

Trident G proposes two linked inversions of established theories of general intelligence and control. Rather than treating adaptive systems as pure free-energy minimisers that drive F toward zero, it treats them as regulators of a defended non-zero set point $F^\star$ within a safe operating band on a cusp, accepting local surprise in exchange for long-run flexibility and metabolic efficiency (Friston, 2010; McEwen, 1998, 2000; Ramstead et al., 2023). And rather than treating g as a single latent capacity that

explains performance, it treats general intelligence as a navigable $\mathcal{G}$ -volume: the portion of cusp state–control space that a brain–body–niche ensemble can reliably reach, traverse, and re-enter without collapse.

On this view, what matters is not just how large an agent’s instantaneous G_f or G_c might be on a given test, but how far the agent can range through Autopilot, Ψ -band, and Trident modes, and how quickly it can return to the defended $F\star$ corridor after perturbation. Capacity measures remain important, but they are folded into a broader control picture in which Ψ -band occupancy, resilience, and niche coupling λ jointly determine how much of the landscape is actually usable for the agent.

Cross-Scale Unification

In its current form, Trident G is deliberately minimal: a single cusp potential over x , with engagement ξ and tilt δ as control dials and a handful of slow meta-states. Its main claim is that this abstract geometry is instantiated by different physical carriers at different scales. At the micro-scale, deviations in excitation–inhibition balance and neuromodulatory tone realise something like x and ξ , with local dopamine balance providing a state-dependent δ -tilt between more convergent and more divergent micro-dynamics (Cools, 2019; Turrigiano, 2012). At the meso-scale, oscillatory rhythms and aperiodic $1/f$ slopes carry the same roles, with beta power, alpha coherence, and gamma spectral tilt jointly tracking movement between Autopilot, Ψ -band, and ‘trident’ wells (Hardstone et al., 2012; Linkenkaer-Hansen et al., 2001; Shine, 2019).

At the macro-scale, large-scale networks and neuromodulatory systems such as LC–NE and dopaminergic midbrain nuclei jointly implement ξ – δ control over fronto-parietal and default-mode configurations, while maintaining near-critical signatures in avalanche exponents, global-signal variability, and heart-rate variability (Aston-Jones & Cohen, 2005; Beggs & Plenz, 2003; Cocchi et al., 2017; Deco & Kringelbach, 2016; Shew & Plenz, 2013; Thayer et al., 2012). On this picture the Ψ -band

is not merely a metaphor but a Griffiths-phase-like region of stretched criticality where general-purpose control tends to live (Moretti & Muñoz, 2013).

Open Questions and Research Programme

The proposal leaves many questions open. We have only sketched how the defended set point $F^\star$ is set and updated over development and across contexts, how Ψ -bands and their boundaries can be reliably detected in vivo, and how best to transform $\mathcal{G}$ into a practical metric for individuals, teams, or AI systems. Empirically, much remains to be done in specifying which combinations of neural, physiological, and behavioural markers could reliably track Ψ -band occupancy, η trajectories, and niche coupling λ , and in testing whether life-span changes in these quantities map onto constructs such as cognitive reserve and neurovisceral integration (Stern, 2009; Thayer et al., 2012).

The accompanying OSF project is intended to turn these sketches into a concrete research programme: it lists candidate tasks, datasets, and analysis pipelines for estimating Ψ -occupancy, $F^\star$ corridors, and $\mathcal{G}$ -volume, and for building Trident G-inspired reinforcement-learning agents. The aim is not to privilege one measure or model class, but to invite competing operationalisations of the same core ideas and to see which ones best predict general problem-solving and resilience in practice.

Life on the Cusp

If Trident G has value, it lies less in claiming a new grand theory of everything and more in offering a compact near-critical control template: a way of thinking about intelligence as life on a cusp. On this view, intelligence is not the absence of surprise, but the capacity to sustain a viable operating band in a changing world – to move fluidly through Autopilot, Ψ -band, and Trident modes, to resist niche capture, and to reshape environments so that future g-loops maintain operational flexibility. How much of the Ψ -band you can reach, how flexibly you can move within it, and how often you can reliably return may together be a more faithful operationalisation of g than any single capacity score.

References

- Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mané, D. (2016). Concrete problems in AI safety (No. arXiv:1606.06565). arXiv.
<https://doi.org/10.48550/arXiv.1606.06565>
- Ashton Smith, M. (2025). Trident G: Computational theory of general intelligence (OSF project). OSF. <https://osf.io/fyt2u/overview>
- Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus-norepinephrine function: Adaptive gain and optimal performance. *Annual Review of Neuroscience*, 28, 403–450. <https://doi.org/10.1146/annurev.neuro.28.061604.135709>
- Barreto, A., Dabney, W., Munos, R., Hunt, J. J., Schaul, T., van Hasselt, H. P., & Silver, D. (2017). Successor features for transfer in reinforcement learning. In I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, & R. Garnett (Eds), *Advances in Neural Information Processing Systems* (Vol. 30). Curran Associates, Inc.
- Beggs, J. M., & Plenz, D. (2003). Neuronal avalanches in neocortical circuits. *Journal of Neuroscience*, 23(35), 11167–11177. <https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003>
- Breton-Provencher, V., & Sur, M. (2019). Active control of arousal by a locus coeruleus GABAergic circuit. *Nature Neuroscience*, 22(2), 218–228. <https://doi.org/10.1038/s41593-018-0305-z>
- Castellanos, F. X., & Proal, E. (2012). Large-scale brain systems in ADHD: Beyond the prefrontal-striatal model. *Trends in Cognitive Sciences*, 16(1), 17–26.
<https://doi.org/10.1016/j.tics.2011.11.007>
- Cocchi, L., Gollo, L. L., Zalesky, A., & Breakspear, M. (2017). Criticality in the brain: A synthesis of neurobiology, models and cognition. *Progress in Neurobiology*, 158, 132–152.
<https://doi.org/10.1016/j.pneurobio.2017.07.002>
- Cools, R. (2019). Chemistry of the adaptive mind: Lessons from dopamine. *Neuron*, 104(1), 113–131. <https://doi.org/10.1016/j.neuron.2019.09.035>

- Critchley, H. D., & Harrison, N. A. (2013). Visceral influences on brain and behavior. *Neuron*, 77(4), 624–638. <https://doi.org/10.1016/j.neuron.2013.02.008>
- Deco, G., & Kringelbach, M. L. (2016). Metastability and coherence: Extending the communication through coherence hypothesis using a whole-brain computational perspective. *Trends in Neurosciences*, 39(3), 125–135. <https://doi.org/10.1016/j.tins.2016.01.001>
- Dixon, M. L., De La Vega, A., Mills, C., Andrews-Hanna, J., Spreng, R. N., Cole, M. W., & Christoff, K. (2018). Heterogeneity within the frontoparietal control network and its relationship to the default and dorsal attention networks. *Proceedings of the National Academy of Sciences*, 115(7), E1598–E1607. <https://doi.org/10.1073/pnas.1715766115>
- Friston, K. (2010). The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138. <https://doi.org/10.1038/nrn2787>
- Gao, R., Peterson, E. J., & Voytek, B. (2017). Inferring synaptic excitation/inhibition balance from field potentials. *NeuroImage*, 158, 70–78. <https://doi.org/10.1016/j.neuroimage.2017.06.078>
- Gershman, S. J. (2018). The successor representation: Its computational logic and neural substrates. *The Journal of Neuroscience*, 38(33), 7193–7200. <https://doi.org/10.1523/JNEUROSCI.0151-18.2018>
- Gillan, C. M., Kosinski, M., Whelan, R., Phelps, E. A., & Daw, N. D. (2016). Characterizing a psychiatric symptom dimension related to deficits in goal-directed control. *eLife*, 5, e11305. <https://doi.org/10.7554/eLife.11305>
- Goldberger, A. L., Amaral, L. A. N., Hausdorff, J. M., Ivanov, P. Ch., Peng, C.-K., & Stanley, H. E. (2002). Fractal dynamics in physiology: Alterations with disease and aging. *Proceedings of the National Academy of Sciences*, 99(suppl_1), 2466–2472. <https://doi.org/10.1073/pnas.012579499>
- Goldstein, D. S., & McEwen, B. (2002). Allostasis, homeostats, and the nature of stress. *Stress*, 5(1), 55–58. <https://doi.org/10.1080/102538902900012345>

- Haldeman, C., & Beggs, J. M. (2005). Critical branching captures activity in living neural networks and maximizes the number of metastable states. *Physical Review Letters*, 94(5), 058101. <https://doi.org/10.1103/PhysRevLett.94.058101>
- Hardstone, R., Poil, S.-S., Schiavone, G., Jansen, R., Nikulin, V. V., Mansvelder, H. D., & Linkenkaer-Hansen, K. (2012). Detrended fluctuation analysis: A scale-free view on neuronal oscillations. *Frontiers in Physiology*, 3. <https://doi.org/10.3389/fphys.2012.00450>
- Henderson, P., Islam, R., Bachman, P., Pineau, J., Precup, D., & Meger, D. (2018). Deep reinforcement learning that matters. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 3207–3214. <https://doi.org/10.1609/aaai.v32i1.11694>
- Hohol, M., Wołoszyn, K., & Brożek, B. (2021). Making cognitive niches explicit: On the importance of external cognitive representations in accounting for cumulative culture. *Frontiers in Integrative Neuroscience*, 15. <https://doi.org/10.3389/fnint.2021.734930>
- Iyengar, N., Peng, C. K., Morin, R., Goldberger, A. L., & Lipsitz, L. A. (1996). Age-related alterations in the fractal scaling of cardiac interbeat interval dynamics. *The American Journal of Physiology*, 271(4 Pt 2), R1078-1084. <https://doi.org/10.1152/ajpregu.1996.271.4.R1078>
- Jepma, M., & Nieuwenhuis, S. (2011). Pupil diameter predicts changes in the exploration–exploitation trade-off: Evidence for the adaptive gain theory. *Journal of Cognitive Neuroscience*, 23(7), 1587–1596. <https://doi.org/10.1162/jocn.2010.21548>
- Kanai, R., Komura, Y., Shipp, S., & Friston, K. (2015). Cerebral hierarchies: Predictive processing, precision and the pulvinar. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 370(1668), 20140169. <https://doi.org/10.1098/rstb.2014.0169>
- Keramati, M., Dezfouli, A., & Piray, P. (2011). Speed/accuracy trade-off between the habitual and the goal-directed processes. *PLOS Computational Biology*, 7(5), e1002055. <https://doi.org/10.1371/journal.pcbi.1002055>

- Khetarpal, K., Riemer, M., Rish, I., & Precup, D. (2022). Towards continual reinforcement learning: A review and perspectives. *Journal of Artificial Intelligence Research*, 75, 1401–1476.
<https://doi.org/10.1613/jair.1.13673>
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., Hassabis, D., Clopath, C., Kumaran, D., & Hadsell, R. (2017). Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13), 3521–3526. <https://doi.org/10.1073/pnas.1611835114>
- Lendner, J. D., Helfrich, R. F., Mander, B. A., Romundstad, L., Lin, J. J., Walker, M. P., Larsson, P. G., & Knight, R. T. (2020). An electrophysiological marker of arousal level in humans. *eLife*, 9, e55092. <https://doi.org/10.7554/eLife.55092>
- Linkenkaer-Hansen, K., Nikouline, V. V., Palva, J. M., & Ilmoniemi, R. J. (2001). Long-range temporal correlations and scaling behavior in human brain oscillations. *Journal of Neuroscience*, 21(4), 1370–1377. <https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001>
- McEwen, B. S. (1998). Stress, adaptation, and disease. Allostasis and allostatic load. *Annals of the New York Academy of Sciences*, 840, 33–44.
<https://doi.org/10.1111/j.1749-6632.1998.tb09546.x>
- McEwen, B. S. (2000). Allostasis and allostatic load: Implications for neuropsychopharmacology. *Neuropsychopharmacology*, 22(2), 108–124. [https://doi.org/10.1016/S0893-133X\(99\)00129-3](https://doi.org/10.1016/S0893-133X(99)00129-3)
- McEwen, B. S., & Gianaros, P. J. (2010). Central role of the brain in stress and adaptation: Links to socioeconomic status, health, and disease. *Annals of the New York Academy of Sciences*, 1186, 190–222. <https://doi.org/10.1111/j.1749-6632.2009.05331.x>
- Menon, V. (2011). Large-scale brain networks and psychopathology: A unifying triple network model. *Trends in Cognitive Sciences*, 15(10), 483–506.
<https://doi.org/10.1016/j.tics.2011.08.003>

- Momennejad, I., Russek, E. M., Cheong, J. H., Botvinick, M. M., Daw, N. D., & Gershman, S. J. (2017). The successor representation in human reinforcement learning. *Nature Human Behaviour*, 1(9), 680–692. <https://doi.org/10.1038/s41562-017-0180-8>
- Moretti, P., & Muñoz, M. A. (2013). Griffiths phases and the stretching of criticality in brain networks. *Nature Communications*, 4(1), 2521. <https://doi.org/10.1038/ncomms3521>
- Palva, J. M., Zhigalov, A., Hirvonen, J., Korhonen, O., Linkenkaer-Hansen, K., & Palva, S. (2013). Neuronal long-range temporal correlations and avalanche dynamics are correlated with behavioral scaling laws. *Proceedings of the National Academy of Sciences*, 110(9), 3585–3590. <https://doi.org/10.1073/pnas.1216855110>
- Peng, C. K., Havlin, S., Stanley, H. E., & Goldberger, A. L. (1995). Quantification of scaling exponents and crossover phenomena in nonstationary heartbeat time series. *Chaos*, 5(1), 82–87. <https://doi.org/10.1063/1.166141>
- Pinker, S. (2010). The cognitive niche: Coevolution of intelligence, sociality, and language. *Proceedings of the National Academy of Sciences*, 107(Suppl. 2), 8993–8999. <https://doi.org/10.1073/pnas.0914630107>
- Ramstead, M. J. D., Sakthivadivel, D. A. R., Heins, C., Koudahl, M., Millidge, B., Da Costa, L., Klein, B., & Friston, K. J. (2023). On Bayesian mechanics: A physics of and by beliefs. *Interface Focus*, 13(3), 20220029. <https://doi.org/10.1098/rsfs.2022.0029>
- Ros, T., Baars, B. J., Lanius, R. A., & Vuilleumier, P. (2014). Tuning pathological brain oscillations with neurofeedback: A systems neuroscience framework. *Frontiers in Human Neuroscience*, 8, 1008. <https://doi.org/10.3389/fnhum.2014.01008>
- Seeley, W. W., Menon, V., Schatzberg, A. F., Keller, J., Glover, G. H., Kenna, H., Reiss, A. L., & Greicius, M. D. (2007). Dissociable intrinsic connectivity networks for salience processing and executive control. *Journal of Neuroscience*, 27(9), 2349–2356. <https://doi.org/10.1523/JNEUROSCI.5587-06.2007>

- Shenhav, A., Botvinick, M. M., & Cohen, J. D. (2013). The expected value of control: An integrative theory of anterior cingulate cortex function. *Neuron*, 79(2), 217–240.
<https://doi.org/10.1016/j.neuron.2013.07.007>
- Shenhav, A., Musslick, S., Lieder, F., Kool, W., Griffiths, T. L., Cohen, J. D., & Botvinick, M. M. (2017). Toward a rational and mechanistic account of mental effort. *Annual Review of Neuroscience*, 40, 99–124. <https://doi.org/10.1146/annurev-neuro-072116-031526>
- Shew, W. L., & Plenz, D. (2013). The functional benefits of criticality in the cortex. *The Neuroscientist: A Review Journal Bringing Neurobiology, Neurology and Psychiatry*, 19(1), 88–100. <https://doi.org/10.1177/1073858412445487>
- Shine, J. M. (2019). Neuromodulatory influences on integration and segregation in the brain. *Trends in Cognitive Sciences*, 23(7), 572–583. <https://doi.org/10.1016/j.tics.2019.04.002>
- Skalse, J., Casper, S., Palencia, M., & Krueger, D. (2022). Defining and characterizing reward hacking. arXiv. <https://doi.org/10.48550/arXiv.2209.13085>
- Srinivasa, N., Stepp, N. D., & Cruz-Albrecht, J. (2015). Criticality as a set-point for adaptive behavior in neuromorphic hardware. *Frontiers in Neuroscience*, 9.
<https://doi.org/10.3389/fnins.2015.00449>
- Stachenfeld, K. L., Botvinick, M. M., & Gershman, S. J. (2017). The hippocampus as a predictive map. *Nature Neuroscience*, 20(11), 1643–1653. <https://doi.org/10.1038/nn.4650>
- Stern, Y. (2009). Cognitive reserve. *Neuropsychologia*, 47(10), 2015–2028.
<https://doi.org/10.1016/j.neuropsychologia.2009.03.004>
- Thayer, J. F., Ahs, F., Fredrikson, M., Sollers, J. J., & Wager, T. D. (2012). A meta-analysis of heart rate variability and neuroimaging studies: Implications for heart rate variability as a marker of stress and health. *Neuroscience and Biobehavioral Reviews*, 36(2), 747–756.
<https://doi.org/10.1016/j.neubiorev.2011.11.009>

Thayer, J. F., & Lane, R. D. (2000). A model of neurovisceral integration in emotion regulation and dysregulation. *Journal of Affective Disorders*, 61(3), 201–216.

[https://doi.org/10.1016/s0165-0327\(00\)00338-4](https://doi.org/10.1016/s0165-0327(00)00338-4)

Thom, R. (1975). *Structural stability and morphogenesis: An outline of a general theory of models* (D. H. Fowler, Trans.). W. A. Benjamin.

Turrigiano, G. (2012). Homeostatic synaptic plasticity: Local and global mechanisms for stabilizing neuronal function. *Cold Spring Harbor Perspectives in Biology*, 4(1), a005736.

<https://doi.org/10.1101/cshperspect.a005736>

Vatansever, D., Menon, D. K., & Stamatakis, E. A. (2017). Default mode contributions to automated information processing. *Proceedings of the National Academy of Sciences*, 114(48),

12821–12826. <https://doi.org/10.1073/pnas.1710521114>

Yin, H. H., & Knowlton, B. J. (2006). The role of the basal ganglia in habit formation. *Nature Reviews Neuroscience*, 7(6), 464–476. <https://doi.org/10.1038/nrn1919>

Zeeman, E. C. (1976). Catastrophe theory. *Scientific American*, 234(4), 65–83.

<https://doi.org/10.1038/scientificamerican0476-65>

Supplementary Material

A companion Open Science Framework (OSF) repository (<https://osf.io/fyt2u/overview>) contains the full computational specification of Trident G. It includes the formal equations for the cusp dynamics and meta-states, code for generating all figures in this article, extended simulations of Ψ -band control and g-loop trajectories, and more detailed task designs and analysis pipelines for the falsifiable predictions outlined below. The aim is for the opinion piece to present the conceptual core, while the OSF project provides a living technical appendix that can be updated as models, benchmarks, and data accumulate.

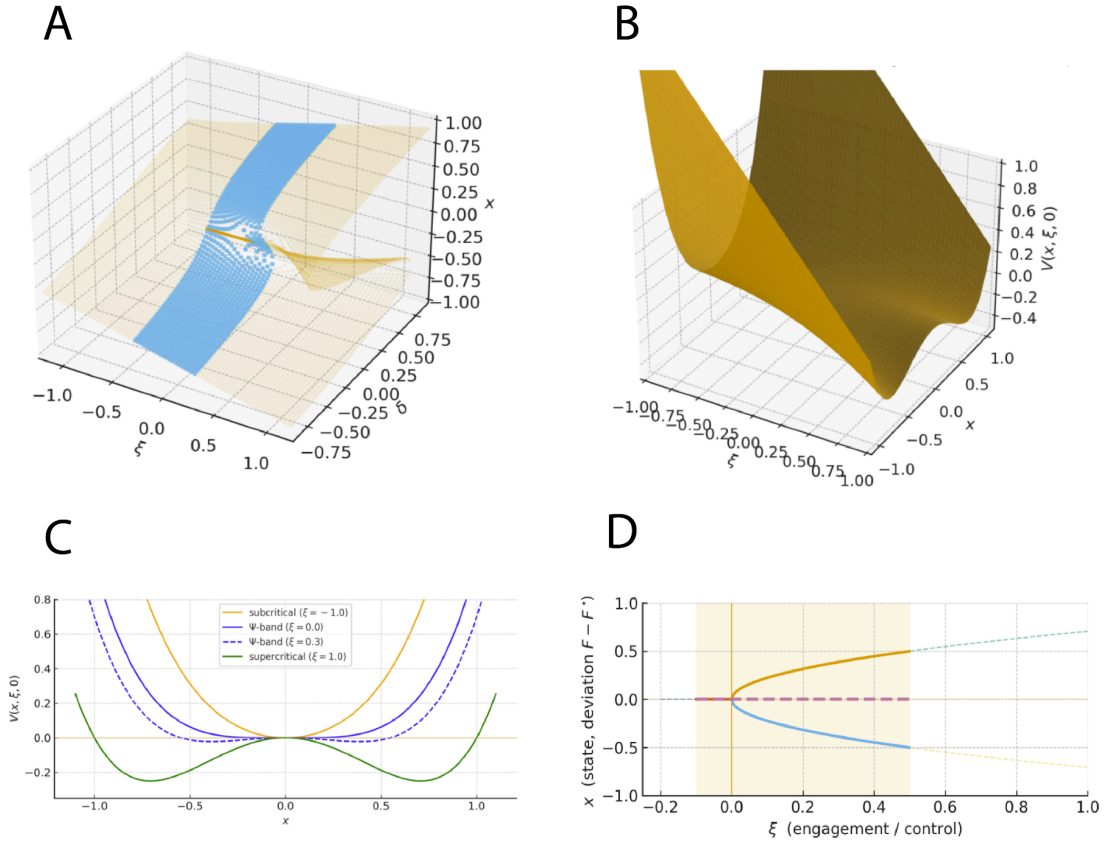


Figure 1. Trident G Cusp Geometry Across Regimes.

A. Equilibrium cusp surface in the (ξ, δ, x) space for the normalised potential $V(x; \xi, \delta) = x^4 - \xi x^2 - \delta x$, with meta-states η, χ, λ held fixed so that ξ can be read as effective engagement. Each point is a stationary state satisfying $\partial V / \partial x = 0$. Grey points show the full equilibrium surface; darker points indicate the Ψ -band collar, here illustrated as the near-critical region $|\xi| \leq \varepsilon$ where the double well is shallow and barrier heights are low. The solid curve shows an example Ψ -band g-loop constrained to $|\xi| \leq \varepsilon$ at zero tilt ($\delta = 0$), running along the central branch near $x \approx 0$, where switching between modes is easy.

B. Normalised cusp potential $V(x; \zeta, 0) = x^4 - \zeta x^2$ at zero tilt, plotted over state deviation $x = F - F^\star$ (horizontal axis) and engagement-like control parameter ζ (horizontal axis). For $\zeta < 0$, the surface has a single central basin (low- $|x|$ Autopilot / Gc -like operation). As $\zeta \rightarrow 0$, this basin flattens into a near-critical Ψ -band region where barriers are low and switching is easy. For $\zeta > 0$, the central point becomes unstable and two outer Gf “bowls” appear at larger $|x|$, corresponding to higher-load, metabolically costly problem-solving states that the system can enter under high engagement.

C. One-dimensional slices of the normalised cusp potential $V(x; \zeta, 0) = x^4 - \zeta x^2$ for representative values of ζ . The horizontal axis is state deviation $x = F - F^\star$; its magnitude $|x|$ measures deviation from the defended set-point (overall cognitive / metabolic load), while the sign simply labels two symmetric branches. For subcritical engagement ($\zeta = -1.0$) the landscape has a single low- $|x|$ well (Autopilot / Gc -like operation). As ζ approaches and then crosses the cusp ($\zeta = 0.0, 0.3$), the central basin flattens and begins to split, marking the Ψ -band where barriers are low and switching between modes is easy. For supercritical engagement ($\zeta = 1.0$), two deep high- $|x|$ wells appear, representing two qualitatively distinct high-load Gf branches (for example, exploration-dominated versus exploitation-dominated problem-solving).

D. Trident projection of equilibrium states in the (ζ, x) plane for the normalised cusp potential $V(x; \zeta, \delta) = x^4 - \zeta x^2 - \delta x$ at zero tilt ($\delta = 0$). The vertical axis is state deviation $x = F - F^\star$; $|x|$ indexes overall cognitive / metabolic load relative to the defended set-point. The lightly shaded band marks an illustrative Ψ -band range of engagement (from $\zeta \approx -0.1$ up to $\zeta = \varepsilon$), capturing the upper end of the Autopilot shaft and the low-barrier Gf regime. For slightly negative engagement the system follows the Autopilot shaft near $x \approx 0$ (low- $|x|$ Gc -like routine control). As ζ crosses zero into the Ψ -band, the equilibrium splits into two high- $|x|$ branches (solid curves), corresponding to exploration-dominated and exploitation-dominated Gf modes. The heavy dashed segment at $x \approx 0$ within the shaded band marks the unstable cusp knife-edge, where small, state-dependent tilts $\delta(t)$ or noise can send processing into one branch or the other. Beyond

the Ψ -band ($\zeta > \varepsilon$), the same branches continue as dashed curves, representing more deeply committed high-load Gf states outside the near-critical corridor.